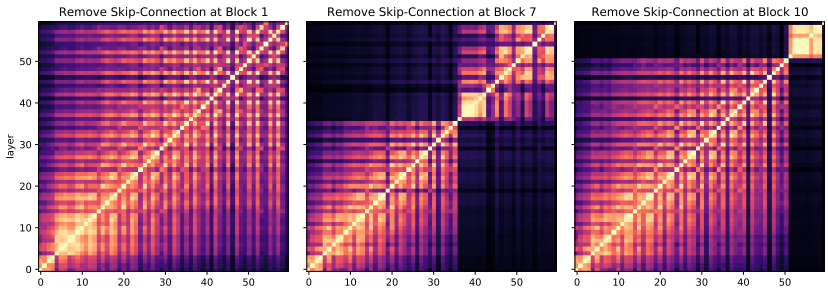


Figure 8: ViT models trained without any skip connections in block i show very little representation similarity between layers before/after block i. We train several ViT models without any skip connections at block i for varying i to interventionally test the effect on representation structure. For middle blocks without skip connections, we observe a performance drop of 4%. We also observe that removing a skip connection at block i partitions similar representations to before/after block i — this demonstrates the importance of skip connections in ViT's standard uniform representation structure.

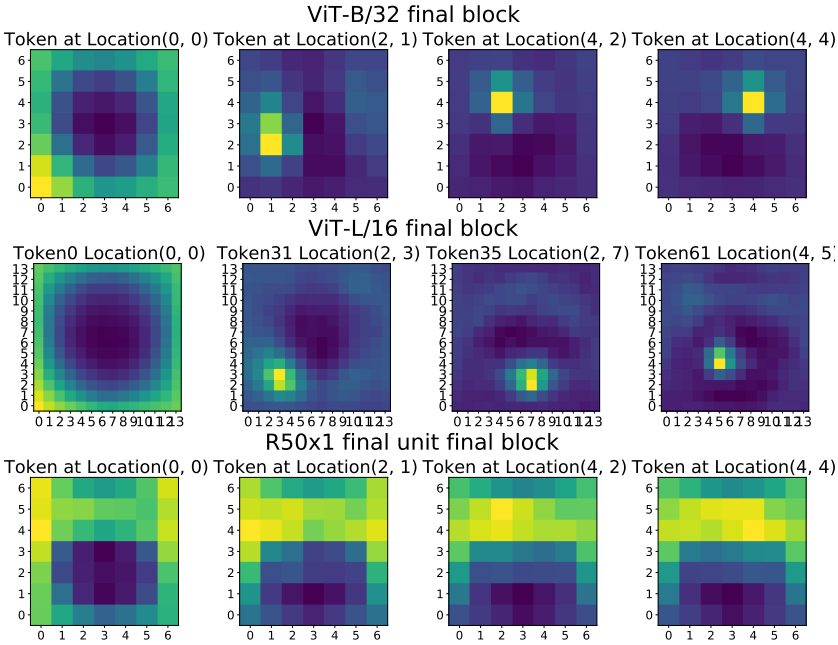


Figure 9: Higher layers of ViT maintain spatial location information more faithfully than ResNets. Each heatmap plot shows the CKA similarity between a single token representation in final block of the model and the input images, which are divided into non-overlapping patches. We observe that ViT tokens have strongest similarity to their corresponding spatial location in the image, but tokens corresponding to spatial locations at the edge of the image (e.g. token 0) additionally show similarity to other edge positions. This demonstrates that spatial information from the input is preserved even at the final layer of ViT. By contrast, ResNet "tokens" (features at a specific spatial location) are much less spatially discriminative, showing comparable similarity across a broad set of input spatial locations. See Appendix for additional layers and results.

removing skip connections from middle blocks.) This demonstrates the importance of representations being propagated by skip connections for the uniform similarity structure of ViT in Figure 1.

7 Spatial Information and Localization

The results so far, on the role of self-attention in aggregating spatial information in ViTs, and skip-connections faithfully propagating representations to higher layers, suggest an important followup question: how well can ViTs perform *spatial localization*? Specifically, is spatial information from the input preserved in the higher layers of ViT? And how does it compare in this aspect to ResNet? An affirmative answer to this is crucial for uses of ViT beyond classification, such as object detection.

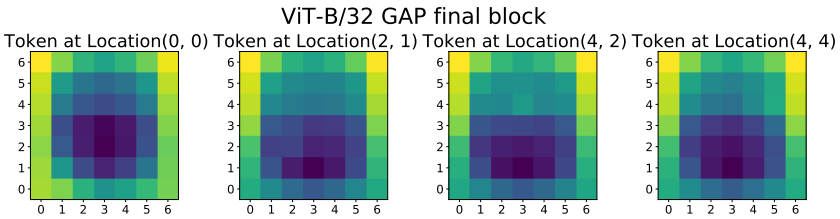
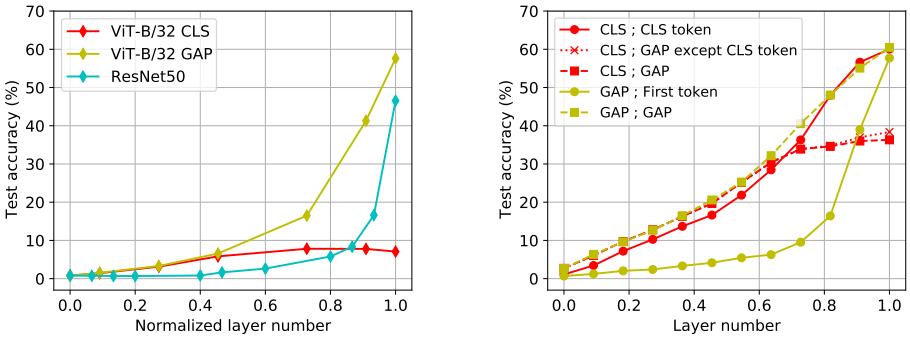


Figure 10: When trained with global average pooling (GAP) instead of a CLS token, ViTs show less clear localization (compare Figure 9). We plot the same CKA heatmap between a token and different input images patches as in Figure 9, but for a ViT model trained with global average pooling (like ResNet) instead of a CLS token. We observe significantly less localization.



(a) Individual token evaluation

(b) CLS vs GAP models

Figure 11: Spatial localization experiments with linear probes. We train linear classifiers on 10-shot ImageNet classification from the representations extracted from different layers of ViT-B/32 models. We then plot the accuracy of the probe versus the (normalized) layer number. **Left:** We train a classifier on each token separately and report the average accuracy over all tokens (excluding the CLS token for the ViT CLS model.) **Right:** Comparison of ViT models pre-trained with a classification token or with global average pooling (GAP) and then evaluated with different ways of aggregating the token representations.

We begin by comparing token representations in the higher layers of ViT and ResNet to those of input patches. Recall that ViT tokens have a corresponding input patch, and thus a corresponding input spatial location. For ResNet, we define a token representation to be all the convolutional channels at a particular spatial location. This also gives it a corresponding input spatial location. We can then take a token representation and compute its CKA score with input image patches at different locations. The results are illustrated for different tokens (with their spatial locations labelled) in Figure 9.

For ViT, we observe that tokens corresponding to locations at the edge of the image are similar to edge image patches, but tokens corresponding to interior locations are well localized, with their representations being most similar to the corresponding image patch. By contrast, for ResNet, we see significantly weaker localization (though Figure D.3 shows improvements for earlier layers.)

One factor influencing this clear difference between architectures is that ResNet is trained to classify with a global average pooling step, while ViT has a separate classification (CLS) token. To examine this further, we test a ViT architecture trained with global average pooling (GAP) for localization (see Appendix A for training details). The results, shown in Figure 10, demonstrate that global average pooling does indeed reduce localization in the higher layers. More results in Appendix Section D.

Localization and Linear Probe Classification: The previous results have looked at localization through direct comparison of each token with input patches. To complete the picture, we look at using each token separately to perform *classification* with linear probes. We do this across different layers of the model, training linear probes to classify image label with closed-form few-shot linear regression similar to Dosovitskiy et al. [14] (details in Appendix A). Results are in Figure 11, with further results in Appendix F. The left pane shows average accuracy of classifiers trained on individual tokens, where we see that ResNet50 and ViT with GAP model tokens perform well at higher layers, while in the standard ViT trained with a CLS token the spatial tokens do poorly – likely because their representations remain spatially localized at higher layers, which makes global classification challenging. Supporting this are results on the right pane, which shows that a single token from the

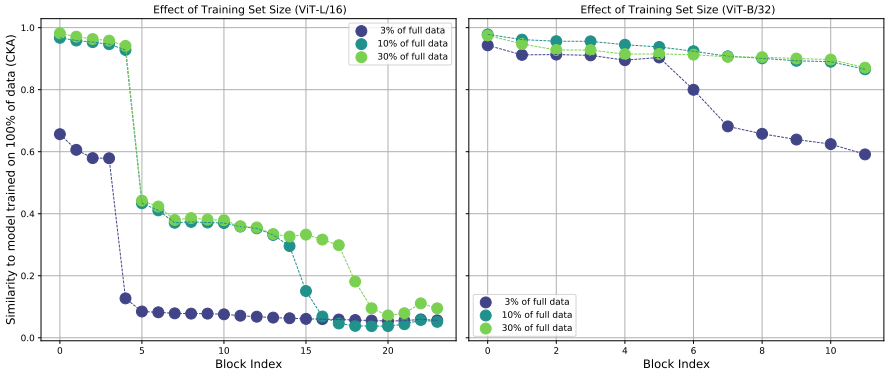
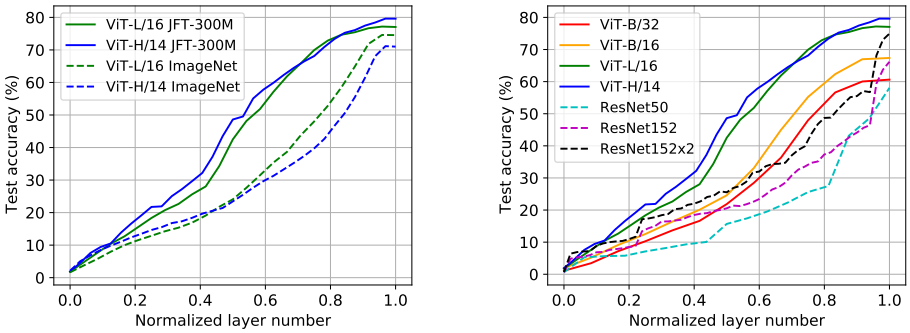


Figure 12: Measuring similarity of representations learned with varying amounts of data shows the importance of large datasets for higher layers and larger model representations. We compute the similarity of representations at each block for ViT models that have been trained on smaller subsets of the data to a model that has been trained on the full data on ViT-L/16 (left) and ViT-B/32 (right). We observe that while lower layer representations have high similarity even with 10% of the data, higher layers and larger models require significantly more data to learn similar representations.



(a) JFT-300M vs ImageNet pre-training

(b) ViTs vs ResNets

Figure 13: Experiments with linear probes. We train linear classifiers on 10-shot ImageNet classification from the aggregated representations of different layers of different models. We then plot the accuracy of the probe versus the (normalized) layer number. **Left:** Comparison of ViTs pre-trained on JFT-300M or ImageNet and evaluated with linear probes on Imagenet. **Right:** Comparison of ViT and ResNet models trained JFT-300m, evaluated with linear probes on ImageNet.

ViT-GAP model achieves comparable accuracy in the highest layer to all tokens pooled together. With the results of Figure 9, this suggests all higher layer tokens in GAP models learn similar (global) representations.

8 Effects of Scale on Transfer Learning

Motivated by the results of Dosovitskiy et al. [14] that demonstrate the importance of dataset scale for high performing ViTs, and our earlier result (Figure 4) on needing scale for local attention, we perform a study of the effect of dataset scale on representations in transfer learning.

We begin by studying the effect on representations as the JFT-300M pretraining dataset size is varied. Figure 12 illustrates the results on ViT-B/32 and ViT-L/16. Even with 3% of the entire dataset, lower layer representations are very similar to the model trained on the whole dataset, but higher layers require larger amounts of pretraining data to learn the same representations as at large data scale, especially with the large model size. In Section G, we study how much representations change in finetuning, finding heterogeneity over datasets.

We next look at dataset size effect on the larger ViT-L/16 and ViT-H/14 models. Specifically, in the left pane of Figure 13, we train linear classifer probes on ImageNet classes for models pretrained on JFT-300M vs models only pretrained on ImageNet. We observe the JFT-300M pretained models achieve much higher accuracies even with middle layer representations, with a 30% gap in absolute